

PROPOSAL FOR EMPLOYEE RETENTION STRATEGY USING MACHINE LEARNING

The IT Consulting Industry Faces a "Talent Crisis"

- High employee turnover is a major cost driver in the IT sector.
- Replacing a skilled consultant costs roughly 6-9 months of their salary (hiring, training, lost productivity).
- **The Problem:** Current retention strategies are "one-size-fits-all" and inefficient.
- **Our Goal:** Move from reactive measures to proactive AI-driven intervention.



Objective: Predict & Prevent High-Value Attrition

INPUT

Historical Employee Data (Demographics, Satisfaction, Income).

ANALYSIS

Identify key drivers of attrition using Machine Learning.

PREDICTION

Detect "High-Risk" employees before they resign.

ACTION

Simulate a "Premium Retention Package" to retain talent profitably.

Utilizing HR Data for Behavioral Analysis

SOURCE

**Company I Internal HR Dataset
(1,470 Employees).**

KEY FEATURES

**Monthly Income, Incentive Levels,
OverTime, Stock Options, Job
Satisfaction.**

PREPROCESSING

**Converted categorical data (Department,
Travel) into numerical format.**

MODELING

**Used LightGBM (Gradient Boosting Machine)
for high accuracy and interpretability.**

VALIDATION

**80/20 Train-Test split to ensure
reliable predictions.**

Attrition is Highly Correlated with Financial Incentives

Observation

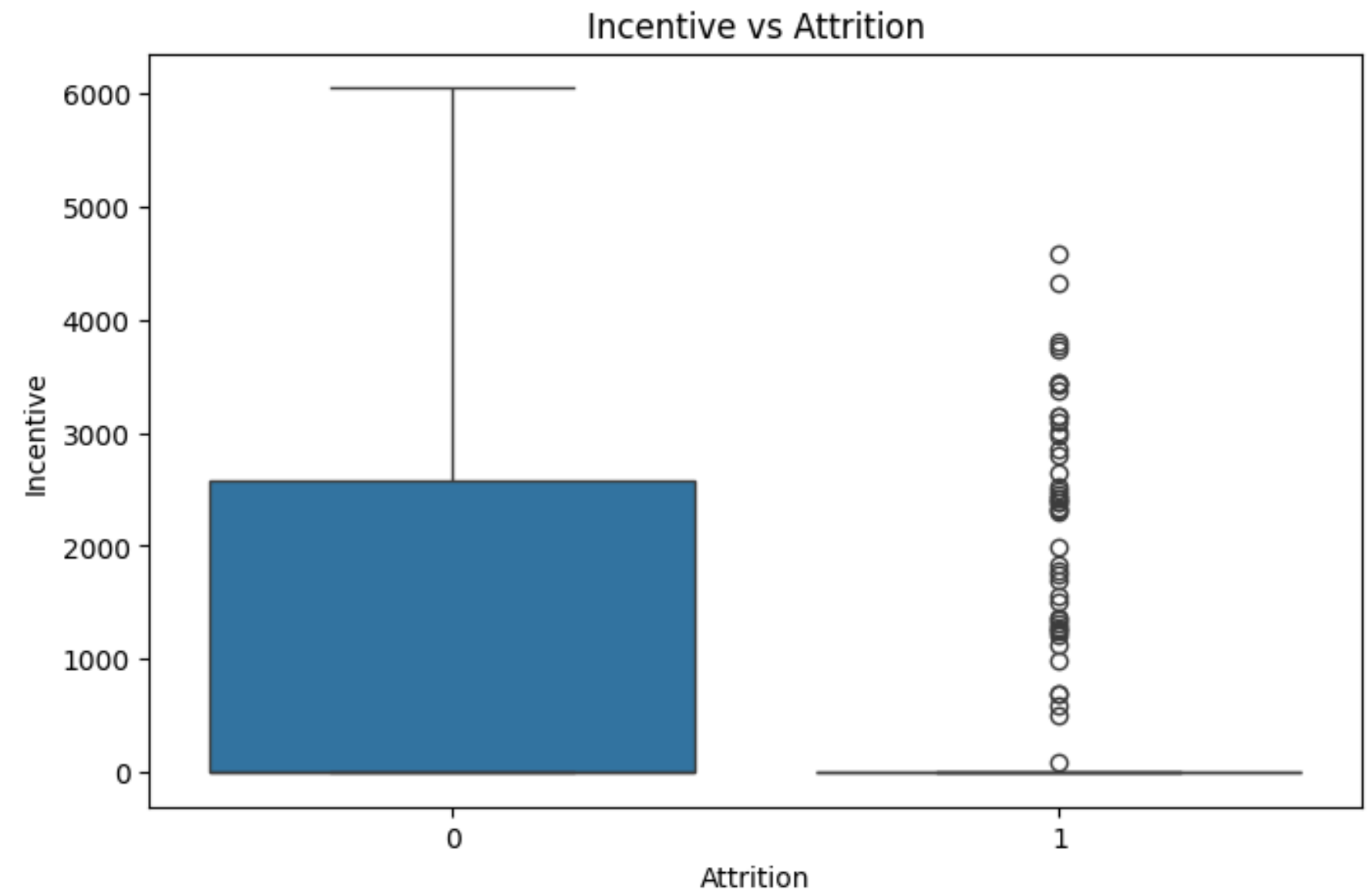
Employees who left (Attrition=1) generally had lower Incentive levels compared to those who stayed.

Correlation

Incentive shows a strong negative correlation (-0.20), meaning as incentives rise, the likelihood of leaving drops.

Conclusion

Financial dissatisfaction is a primary trigger for resignation at Company I.



High-Precision Prediction Using LightGBM

MODEL USED

LightGBM (Gradient Boosting Framework)

MODEL SCORE

0.8486 (Outstanding predictive power)

EVALUATION METRIC

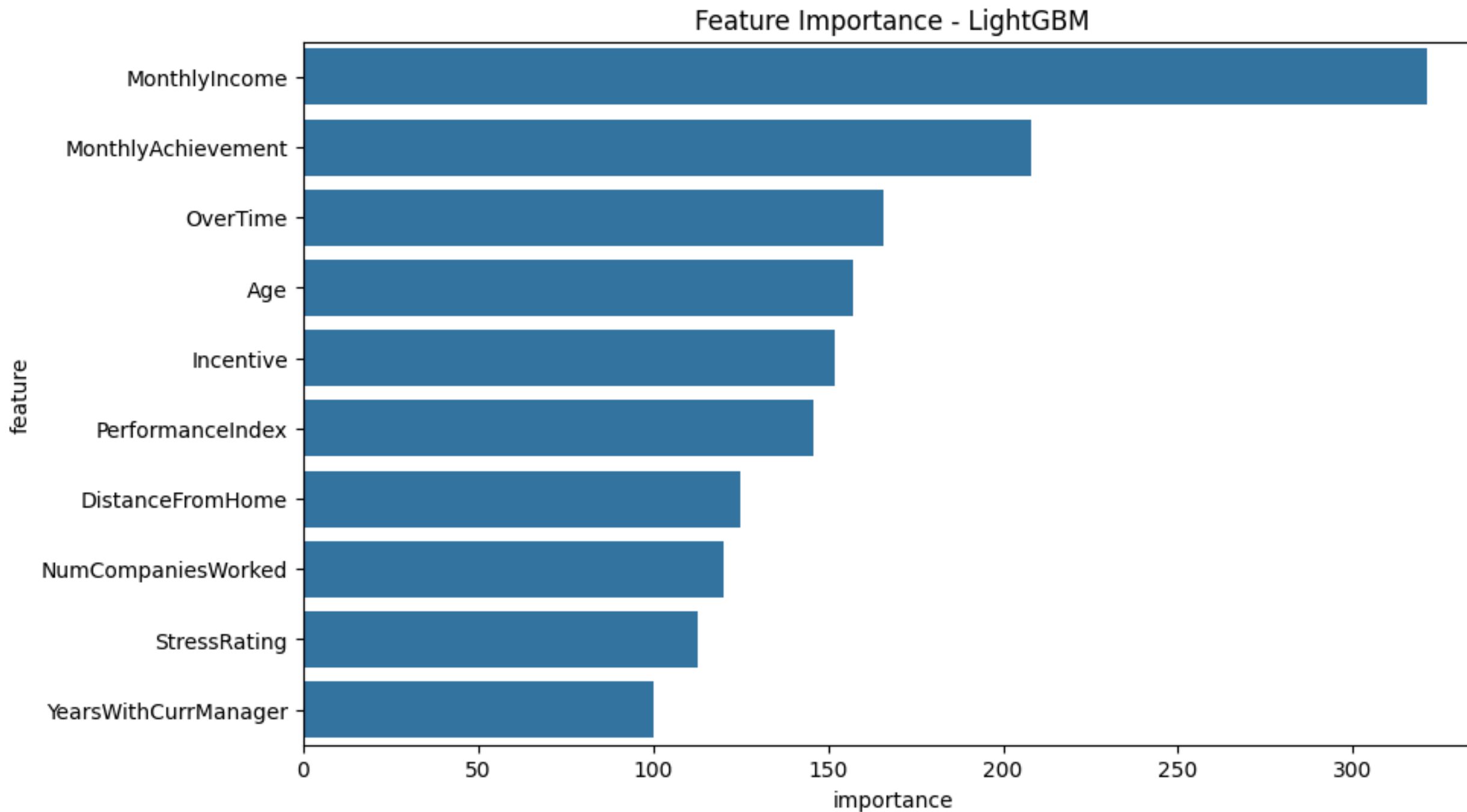
ROC-AUC (Area Under the Curve)

ACCURACY

88.1%

A ROC-AUC score above 0.8 indicating the model is excellent at distinguishing between employees who will stay and those who will leave.

What Matters Most to Employees?



Top Predictors

- **Monthly Income:** The strongest indicator of retention.
- **OverTime:** A key indicator of burnout risk.
- **Incentive / Stock Options:** Critical levers for retention.

Business Implication

We cannot easily reduce "OverTime" immediately, but we can adjust "Income/Incentives" to retain key talent.

Proposing a "Targeted Retention System"

The Flaw in Current Systems: Giving raises to everyone is too expensive. Ignoring the problem is even costlier.

Our Solution: **A Targeted Intervention**

Mechanism:

- Identify "**High Risk**" employees using the LightGBM model.
- Offer a "**Premium Retention Package**" (Incentive + Stock Option Boost) only to these specific individuals.

Hypothesis: This targeted approach will save significantly more money in avoided hiring costs than it costs in additional incentives.



Projecting the ROI of the "Premium Retention Package"

SCENARIO

applied the intervention to **28 High-Risk Employees** identified by the AI.

COST OF INTERVENTION

\$140,000 (28 employees × \$5,000 package)

SAVINGS GENERATED

\$360,000 (12 employees saved × \$30,000 hiring cost)

NET PROFIT

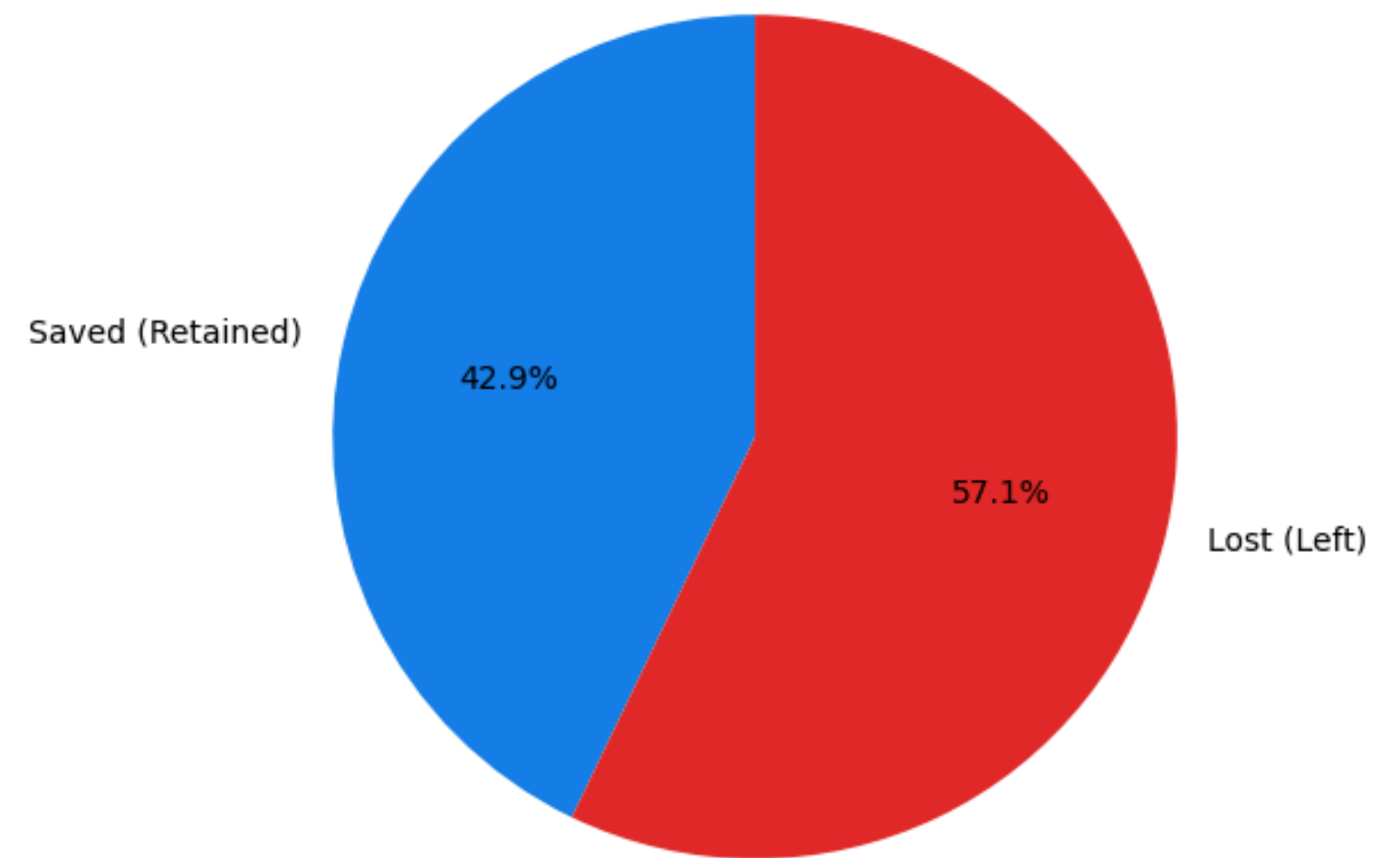
\$220,000

This projection assumes a conservative replacement cost of \$30k per consultant. The actual ROI may be higher for senior roles.

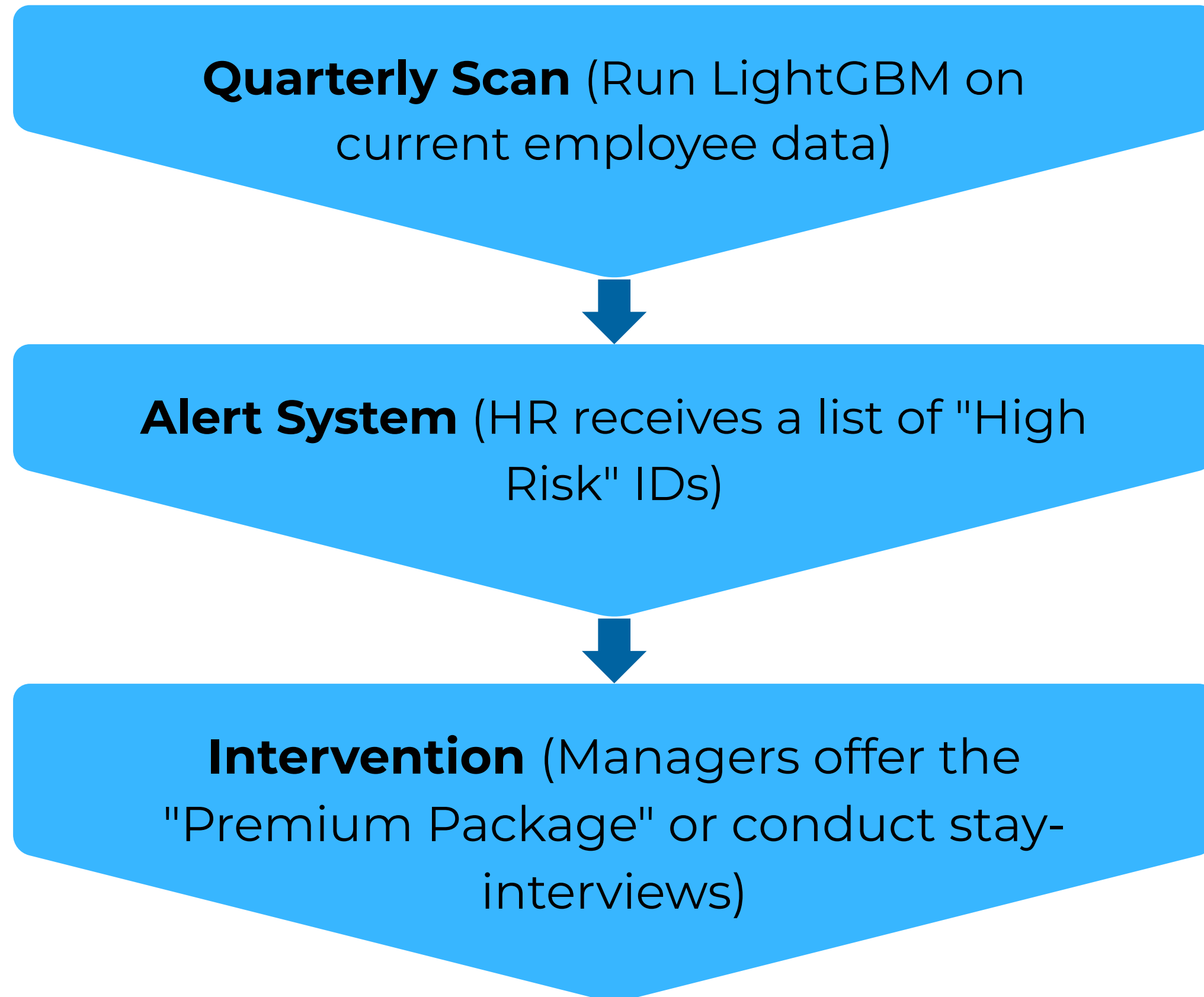
Effectiveness of the Intervention

- **Before Intervention:** 28 employees were predicted to leave with >50% probability.
- **After Intervention:** The model predicts **12 of them (42.9%)** will choose to stay.
- **Insight:** While financial incentives don't solve every case, saving nearly half of the high-risk talent yields massive returns.

Outcome of Targeted Retention Program
(28 High-Risk Employees)



Operationalizing the AI Model



Long-Term Strategy for Company

SHORT TERM

Implement the "Premium Package" for immediate risk reduction.

MEDIUM TERM

Review "OverTime" policies (the #2 predictor) to reduce burnout.

LONG TERM

**Adjust base salary structures to align with market rates, as
MonthlyIncome was the #1 predictor.**

Summary

- We successfully built a predictive model (**AUC 0.85**) to identify attrition risk.
 - Data confirms that **Financial Incentives** and **Stock Options** are the strongest levers for retention.
 - Our "Targeted Retention" simulation demonstrates a potential **Net Profit of \$220,000** by focusing resources only on high-risk staff.
 - **Final Verdict:** AI-driven HR management is not just feasible; it is highly profitable.
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